[METRIC]
A-A-52508A
November 18, 1996
SUPERSEDING
A-A-52508
November 21, 1994

COMMERCIAL ITEM DESCRIPTION

INDICATOR, VOLTAGE: BATTERY-GENERATOR, 28 VOLT DC (NOMINAL) (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers a non-luminous battery-generator voltage indicator (voltmeter) for 28 volt (V) (nominal) direct current (dc) electrical systems.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with the manufacturer's material specifications for voltage indicators.

2.1.1 Cover window material. The cover window material shall be transparent heat resistant acrylic resin based sheet or any transparent plastic molding material which conforms to the salient characteristics specified herein. The window thickness shall be 3.18 $\pm$ 0.38 millimeters (mm).

2.2 Design and construction. The indicator shall be designed and constructed to the form and dimensions specified herein (see figures 1 and 2).

2.2.1 Locking devices. Lockwashers, self-locking nuts, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 6625

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2.2.2 Dial marking. The face of the dial shall be marked as shown in figure 1.

2.2.3 Dial fastening. The dial shall be secured to the indicator body in such a position that the marking shall be legible when illuminated through the case window from an outside source. The method of securing the dial shall be such that it shall not become loose or turn when subject to vibration (see 2.4.1) or shock (see 2.4.2).

2.2.4 Dial illumination. The indicator shall provide side windows or other provisions for adequate indirect illumination of dial in accordance with figure 1.

2.2.5 Pointer. The pointer shall be damped to prevent oscillation of no more than 2 degrees at all indicated points within the range of the indicator, when subjected to vibration (see 2.4.1) and shock (see 2.4.2).

2.2.6 Cover window. The cover window shall be secured to the case so as to prevent moisture from entering the case. The window shall demonstrate an Izod strength of not less than 16 joules (J) per meter of notch when tested as specified by procedure C of ASTM D256.

2.3 Performance.

2.3.1 Calibration. When tested with the indicator mounted in a vertical position, calibration shall be as follows:

a. With the pointer positioned midway between the left and right limits of the indentation in the top of the green section of the generator arc, the applied voltage shall be not less than 28 V nor more than 29 V, except that after testing as specified in 2.3.2, 2.3.3, 2.4.1, and 2.4.2, the applied voltage shall be not less than 27.5 V nor more than 29.5 V.
b. With the pointer positioned at the midpoint between the low voltage red and the yellow sections of the battery arc, the applied voltage shall be not less than 21 V nor more than 23 V, except that after testing as specified in 2.3.2, 2.3.3, 2.4.1, and 2.4.2, the applied voltage shall be not less than 20 V nor more than 24 V.

2.3.2 Temperature influence. The indicator shall operate throughout an ambient air temperature range of minus (-) 54 degrees Celsius (°C) to plus (+) 71°C. To determine compliance, the temperature shall be stabilized at each extreme temperature and readings taken after 4 hours exposure at each extreme temperature. In addition, the indicator shall conform to 2.3.1.

2.3.3 Endurance. When mounted and calibrated as specified in 2.3.1, and after the calibration, the indicator pointer shall be subjected to full scale deflection and return for a total of 200 000 cycles. Deflection shall be caused by application of maximum voltage, utilizing an off-on switch. One cycle of operation shall consist of one complete oscillation from extreme left to extreme right and return to extreme left. Following testing, the indicator shall conform to 2.3.1.

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2.3.4 Dielectric withstanding voltage. If metal case construction is specified, the insulation of the indicator shall show no evidence of cracking or charring when subjected to 440 V root mean square (rms) at a frequency of 60 cycles per second (cps), applied between the case and ungrounded terminal or winding for no less than one minute. This test is to be conducted prior to final assembly for sealed items.

2.3.5 Electrostatic susceptibility. The indicator shall demonstrate no susceptibility to static charge generation. To determine conformance, the indicator shall be conditioned at 23 ± 3°C and 15 percent (%) relative humidity for a minimum of 8 hours. Immediately following the conditioning, the window shall be rubbed with a cloth containing not less than 95% fleece or pulled wool. The pointer shall not deflect from a static charge generated when the window is rubbed.

2.4 Environmental.

2.4.1 Vibration. The indicator shall meet the requirement of 2.2.5 when subjected to a simple harmonic motion having an amplitude of 0.76 mm (1.5 mm maximum total excursion), the frequency being varied uniformly between the approximate limits of 10 and 55 hertz (Hz). The entire frequency range, from 10 to 55 Hz and return to 10 Hz, shall be traversed in approximately 1 minute. This motion shall be applied for a period of two hours in each of three mutually perpendicular directions (total of 6 hours). Following the vibration, the indicator shall be checked for looseness and turning of dial and then shall meet the requirements of 2.3.1.

2.4.2 Shock. The indicator, mounted to simulate actual mounting in use, shall not be damaged and shall meet the requirements of 2.2.5 following exposure to ten shock waves applied to the vertical axis with the indicator in its normal mounting orientation. The test pulse shall be as specified in figure 3. The pulse shall have a duration of 6 milliseconds and a peak value of 100 gravity units (g). An optional method of mechanically inducing shock may be used in such a manner that an impact of 50g will be obtained.

2.4.3 Waterproofness. The indicator shall evidence no leakage and shall be mechanically and electrically operable during and subsequent to submersion when subjected to the following procedure:

a. Prior to submersion, the indicator shall be installed in a test circuit equivalent to its normal environment and operated for a period of 30 minutes at full rated current and voltage.
b. The indicator with its electrical connections shall be submerged in a container of 5% saline solution with the indicator uppermost surface a minimum of 25.4 mm below the surface and with the container in an evacuation chamber. The indicator shall be carefully observed during its entire period of submersion and shall be operated while submerged for 30 minutes at full rated current and voltage. The chamber shall be

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evacuated to a pressure 4183 kilograms per square meter (kg/m^2) below atmospheric pressure so as to apply a minimum of 41 kilopascals (kPa) internal pressure to all voids within the indicator.

c. The chamber shall then be pressurized to 4183 kg/m^2 above atmospheric pressure and the indicator again operated for 30 minutes. Test results shall be compared with the results from 2.4.3a.

2.4.4 Salt fog (corrosion). The indicator shall operate with no evidence of malfunction or failure after 200 hours exposure to a salt-fog environment in accordance with ASTM B117. Following the salt-fog exposure, the indicator shall meet the requirements of 2.3.1.

2.4.5 Fungus. The indicator shall show no evidence of fungus growth affecting operation after exposure to conditions favorable to fungus growth for 90 days. Following the fungus exposure, the indicator shall meet the requirements of 2.3.1.

2.5 Finish.

2.5.1 Bezel. The indicator bezel shall be lusterless forest green color 383 in accordance with Army drawing number 12369000.

2.5.2 Pointer. The visible part of the pointer shall be all white and non-luminous.

2.5.3 Preparation and painting. Cleaning, pretreating, priming, and painting shall conform to an appropriate paint system selected from Army drawing number 12369000. In no case shall paint or other coating be applied to that portion of the case body so as to interfere with the intended purpose of dial illumination (see 2.2.4).

2.5.4 Identification and marking. Identification and marking shall be permanent and legible and shall include as a minimum, the manufacturer's identification code (CAGE), the part identification number (PIN) AA52508-1, the date of manufacture (month, year e.g. April 94), and "US".

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered material to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

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4.2 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 6.2).

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Addresses for obtaining copies of referenced documents.

6.1.1 Army drawings. Copies of Army drawing number 12369000 "Chemical Agent Resistant Coatings (CARC) Paint Systems Index" are available from the U.S. Army Tank-automotive and Armaments Command, Warren, MI 48397-5000.

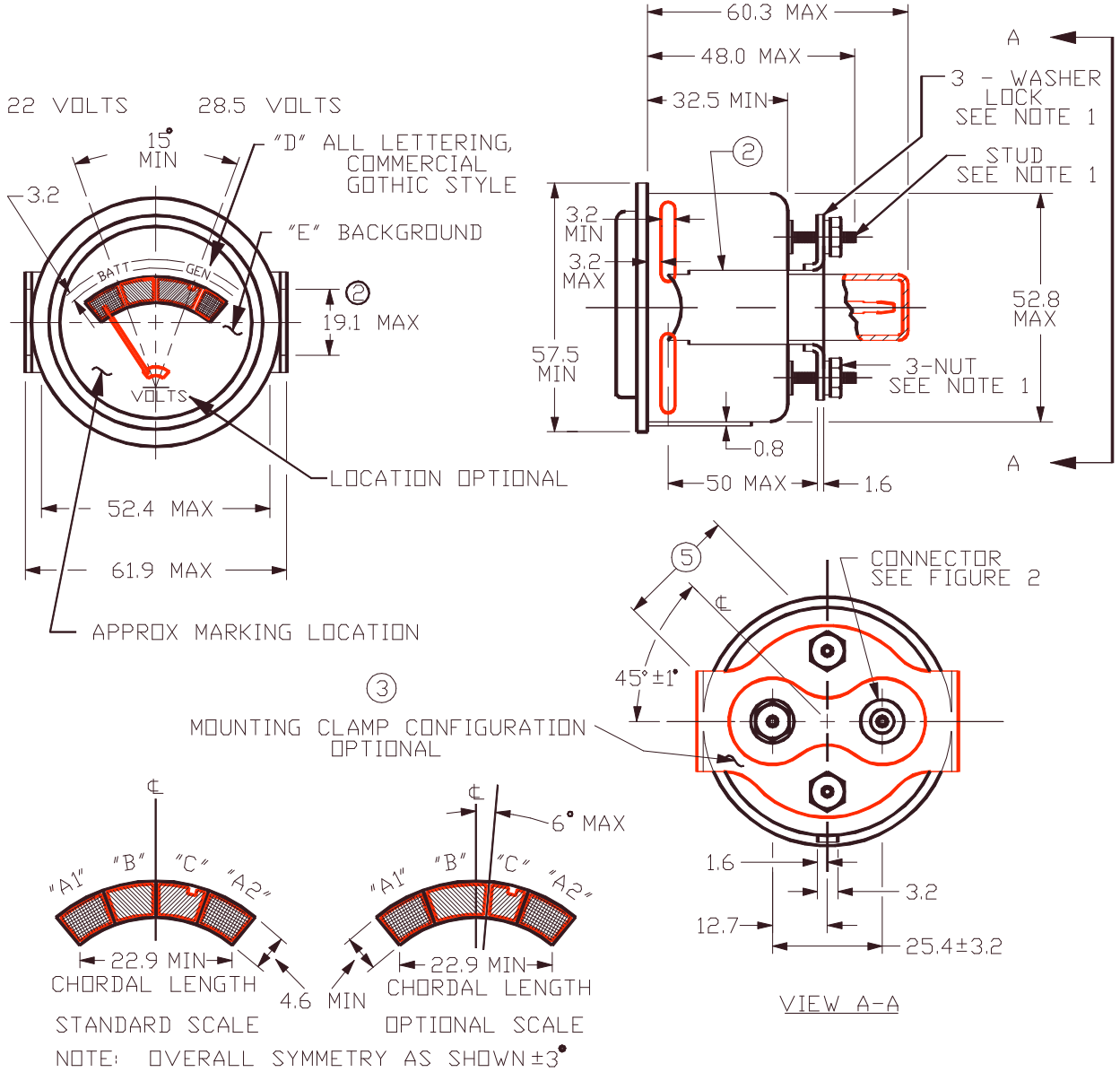
6.1.2 Non-Government publications. ASTM B117 "Standard Practice for Operating Salt Spray (Fog) Apparatus" (DoD Adopted), ASTM D256 "Standard Test Methods for Determining the Pendulum Impact Resistance of Notched Specimens of Plastic" (DoD Adopted), ASTM B700 "Standard Specification for Electrodeposited Coating of Silver for Engineering Use", and ASTM D2000 "Standard Classification System for Rubber Products in Automotive Applications (SAE Recommended Practice J200)" (DoD Adopted) can be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. The Metals and Alloys in the Unified Numbering System (UNS) publication is available from the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Drive, Warrendale, PA 15096.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. PIN and quantity required.
d. Selection of applicable level and packaging requirements.

6.3 Cross-reference data. Indicators conforming to this CID are interchangeable/substitutable with indicators conforming to MIL-I-26525F and MS24532J.

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CODE	COLOR	VOLTAGE RANGE
A1	RED	18 - 22
A2	RED	30 - 34
B	YELLOW	22 - 26
C	GREEN	26 - 30
D	WHITE	
E	BLACK	

CID PART NO.	FORMER MS PART NO.	DIAL TYPE	FORMER PART NO.
———	MS24532-1	CANCELLED (SEE NOTE 8)	———
AA52508-1	MS24532-2	NON-LUMINOUS	8376488

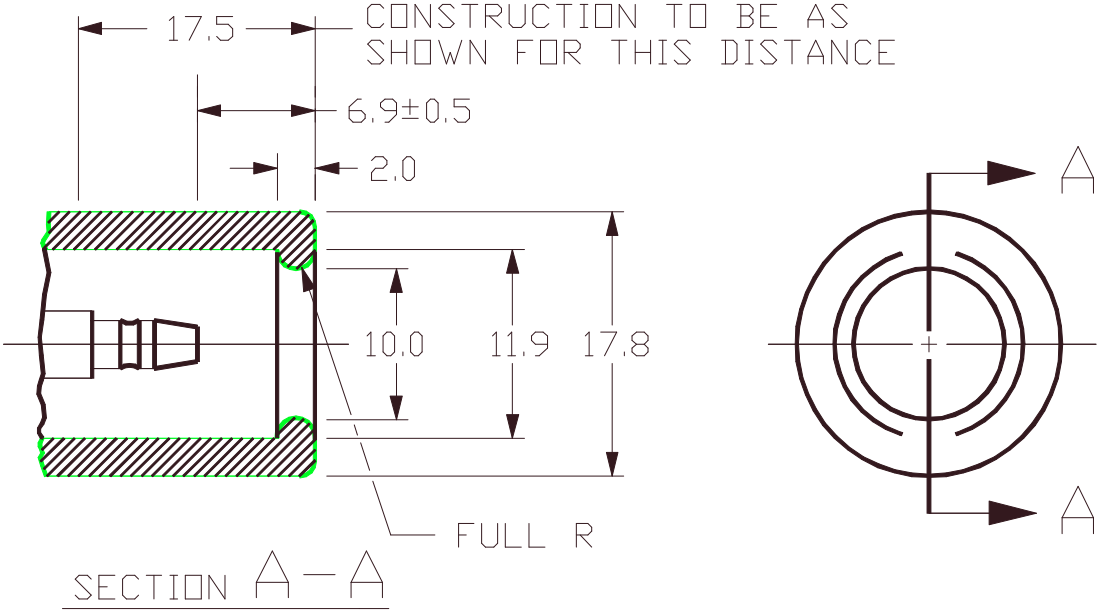
Figure 1. Voltage regulator design, construction, and dimensions.

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NOTES:

1. Thread size of mounting studs, lockwashers, and nuts should be in accordance with the manufacturer's specification for voltage indicators.
2. Mounting clamp tips should be located as shown.
3. Relative location of connectors and mounting studs should be as shown. Location relative to instrument case is optional.
4. Mounting clamp tips should be sharp to assure grounding.
5. Four (4) illuminating windows, 29.4 minimum (min) chordal length, equally spaced 45 degrees (°) off horizontal centerline (360° illuminating window optional).
6. Dimensions are in mm. Unless otherwise specified, tolerances are ± 0.4 mm.
7. This CID is not intended to limit construction other than as shown herein by dimensions, notations, and referenced documents.
8. Part number AA52508-1 replaces part number MS24532-2.

Figure 1. Voltage regulator design, construction, and dimensions - Continued.



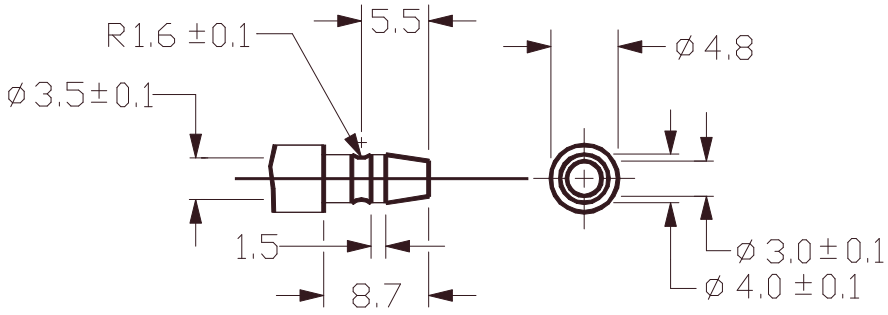


Figure 2. Connector.

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NOTES:

1. Dimensions are in mm.
2. Unless otherwise specified, tolerances are ± 0.1 mm on decimals, ± 1° on angles, and ± 0.4 mm for rubber products.
3. Material:
   a. Shell: Rubber, Grade SC615, A1 C1 E3 F2, in accordance with ASTM D2000.
   b. Terminal: Brass, Unified Numbering System (UNS) No. C26000-C28000, C33000-C34200, C35000-C36000, C37000, and C46400-C48500, temper halfhard minimum.
4. Protective Finish: Terminal should be silver plated to a minimum thickness of 0.006 mm in accordance with ASTM B700, Type I, II, or III, Grade D, Class S.
5. When conductors are coupled, the voltage drop should not exceed 20 ± 1 millivolt with a measurement error of 5% when operated in a 28 V dc circuit.

Figure 2. Connector - Continued.

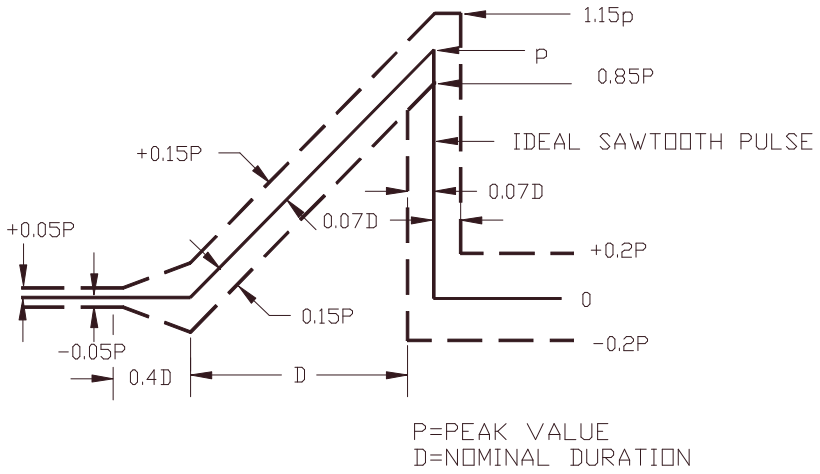


FIGURE 3. Shock pulse.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - MC
Air Force - 99

Review Activities:
Army - CR
Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - AT

(Project 6625-0859)

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